Ramsey's theorem for pairs preserves well-foundedness for the induction and collection schemes

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Editorial note. The counted unit is the article's own Corollary cor:rt22-wf at source lines 1384-1391 together with the proof environment that begins on the next line (1392-1401); the pairing is the ordinary adjacent one.

Editorial note. The retained context is the statement of Proposition prop:rt22-wf at source lines 1356-1363, which the counted proof applies; that proposition is counted by a sibling card of this lane, its proof is not retained, and the counted intervals are disjoint.

Editorial note. The counted proof carries the author citation of Belanger inside its first line. The author's own bibliography file biblio.bib from the pinned e-print of arXiv:2402.11616v3 is attached to this task unedited and complete, with the bibtex backend, so that the citation resolves.

Editorial note. The article has an earlier commented-out version of this same corollary at source lines 1372-1382; it is commented out by the author and is not author text for our purposes, and it is not retained.

Editorial note. The retained preamble reproduces verbatim the author's document class at source line 2, his theorem declarations at source lines 16-25, his macro definitions at source lines 46-88 and 91, and his natural-operation macros at source lines 104-107, and it retains mathabx and cleveref; the stix-dependent LS2 block and the inputenc load with the utf8x option are not retained.

Editorial note. The single bibliography entry needed by this card is provided as a task-directory fragment holding the entry exactly as the bibtex run had formatted it from the author own biblio.bib; the bibtex backend is not used because the harness order is xelatex twice and then bibtex, so a generated bbl is never read back and the citation would stay undefined in the final pass.

Allowed standard prerequisites (editorial).

- The RT22 proposition controlling the jump. For a countable model of RCA_0 plus collection for Sigma^0_2 plus well-foundedness of omega_4 to the alpha, topped by a set Y, and every colouring and every set P

above the jump of Y with the model of P satisfying RCA_0-star, there is a homogeneous set G with P above the jump of G plus Y and with the model of G satisfying collection and well-foundedness of alpha; retained with this card at source lines 1356-1363.. The counted corollary obtains its auxiliary set P and then applies exactly this proposition; the corollary differs from it by replacing collection with an arbitrary induction or collection scheme at a finite level and by dropping the auxiliary set.

- Omega-extension for finite levels of induction and collection. The proof invokes a result attributed to Harrington, Simpson-Smith and Hajek, recorded by Belanger, that every countable topped model of RCA_0-star plus the scheme at level n minus one can be omega-extended into a model of the same theory plus weak Konig's lemma; the citation is carried in the proof and its entry is attached with this card.. It produces the set P above the jump of Y whose model satisfies the scheme at level n minus one, which is what the RT22 proposition needs.

- Transfer of a finite-level scheme along an omega-extension. If a Sigma^0_n formula holds in the extension of G and every such formula is Sigma^0_{n-1} in the model of the auxiliary set P, then satisfaction of the scheme at level n minus one in that model gives satisfaction of the scheme at level n in the model of G; the proof states this transfer in its last two lines.. It is the step that raises the scheme from level n minus one to level n, which is the difference between the hypothesis and the conclusion of the counted corollary.

Proposition 0.1. *Consider a countable model $\mathcal{M} = (M, S) \models \mathsf{RCA}_0 + \mathsf{B}\Sigma^0_2 + \mathsf{WF}(\omega_4^\alpha)$ topped by a set $Y \in S$, where $\alpha \in M$ is an ordinal $\leqslant \epsilon_0$. Then, for every coloring $f : [M]^2 \to 2$ in S and every set $P \gg Y'$ such that $\mathcal{M}[P] \models \mathsf{RCA}^*_0$, there exists $G \subseteq M$ such that*

1. *G is an M-infinite f-homogeneous set ;*
2. *$P \geqslant_T (G \oplus Y)'$;*
3. *$\mathcal{M}[G] \models \mathsf{RCA}_0 + \mathsf{B}\Sigma^0_2 + \mathsf{WF}(\alpha)$.*

Corollary 0.2. *Fix $n \geqslant 2$. Let T_n be either $\mathsf{B}\Sigma^0_n$ or $\mathsf{I}\Sigma^0_n$. Consider a countable model $\mathcal{M} = (M, S) \models \mathsf{RCA}_0 + T_n + \mathsf{WF}(\omega_4^\alpha)$ topped by a set $Y \in S$, where $\alpha \in M$ is an ordinal $\leqslant \epsilon_0$. Then, for every coloring $f : [M]^2 \to 2$ in S, there exists $G \subseteq M$ such that*

1. *G is an M-infinite f-homogeneous set ;*
2. *$\mathcal{M}[G] \models \mathsf{RCA}_0 + T_n + \mathsf{WF}(\alpha)$.*

Proof. By Harrington; Simpson–Smith; Hajek (see Belanger [1]), every countable topped model of $\mathsf{RCA}^*_0 + T_{n-1}$ can be ω-extended into a model of $\mathsf{RCA}^*_0 +$

$T_{n-1} + \mathsf{WKL}$. Since $\mathcal{M}[Y'] \models \mathsf{RCA}_0^* + T_{n-1}$, there is some $P \subseteq M$ such that $P \gg Y'$ and $\mathcal{M}[P] \models \mathsf{RCA}_0^* + T_{n-1}$. By Proposition 0.1, there exists $G \subseteq M$ such that

1. G is an M-infinite f-homogeneous set ;
2. $P \geqslant_T (G \oplus Y)'$;
3. $\mathcal{M}[G] \models \mathsf{RCA}_0 + \mathsf{B}\Sigma^0_2 + \mathsf{WF}(\alpha)$.

Since every Σ^0_n formula in $\mathcal{M}[G]$ is Σ^0_{n-1} in $\mathcal{M}[P]$ and $\mathcal{M}[P] \models T_{n-1}$, $\mathcal{M}[G] \models T_n$. □

References

[1] David R. Bélanger. Conservation theorems for the cohesiveness principle. 2022. Available at `https://arxiv.org/abs/2212.13011`.